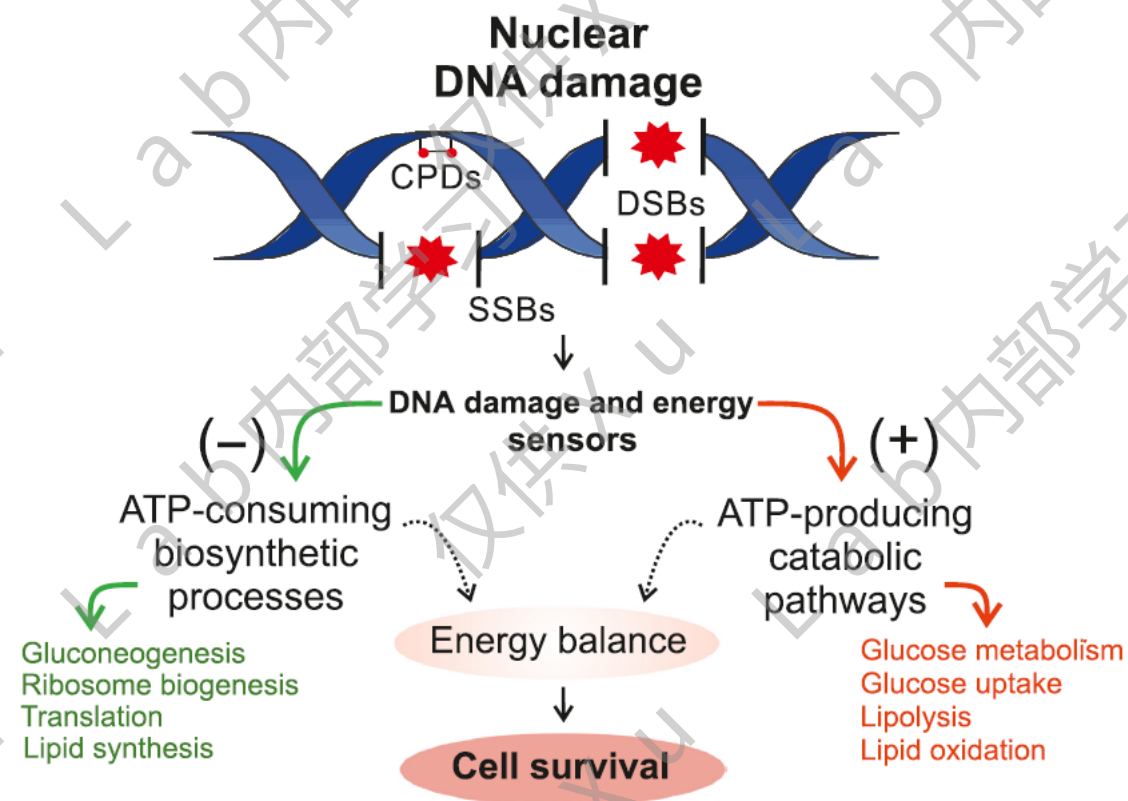
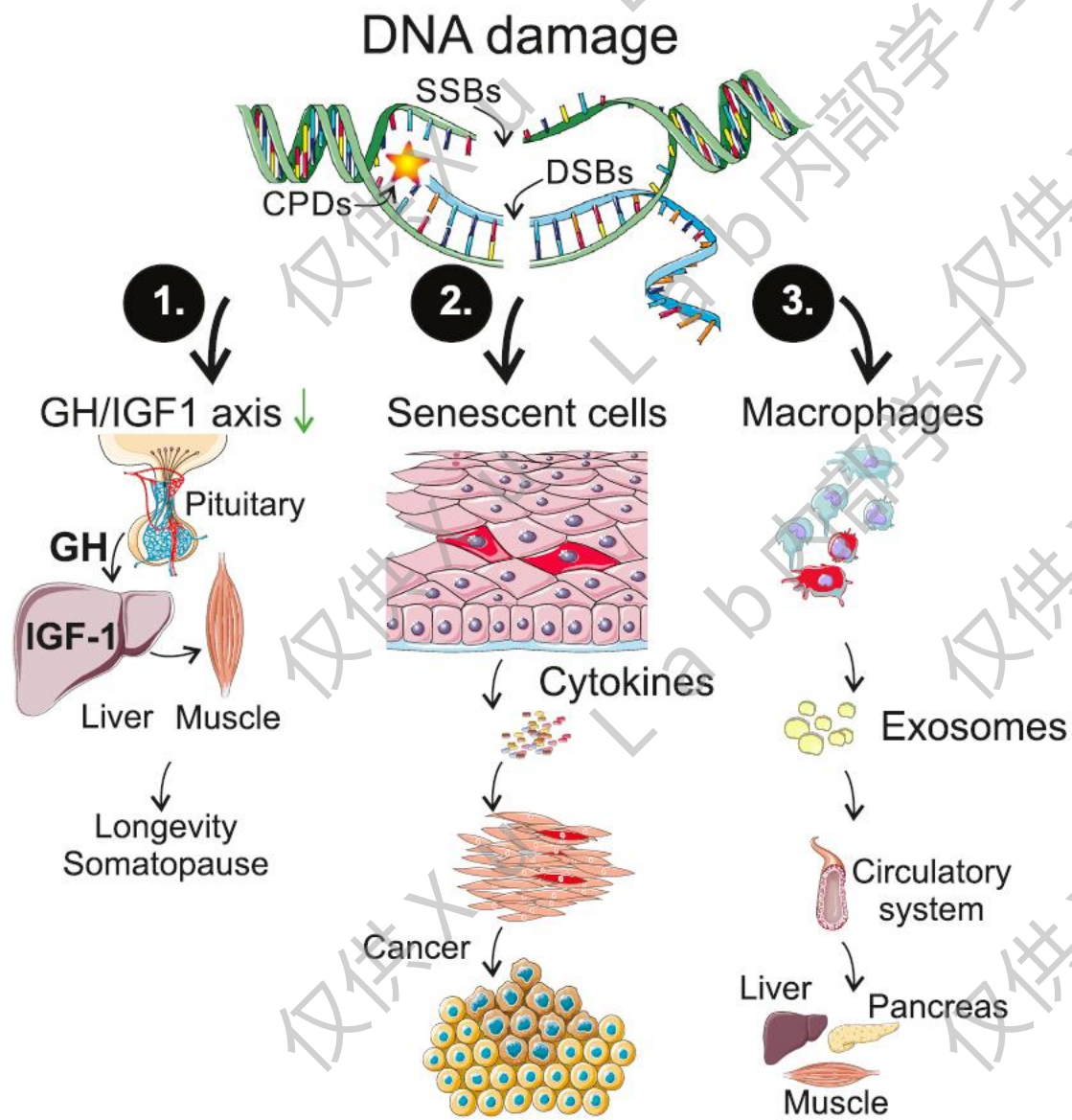
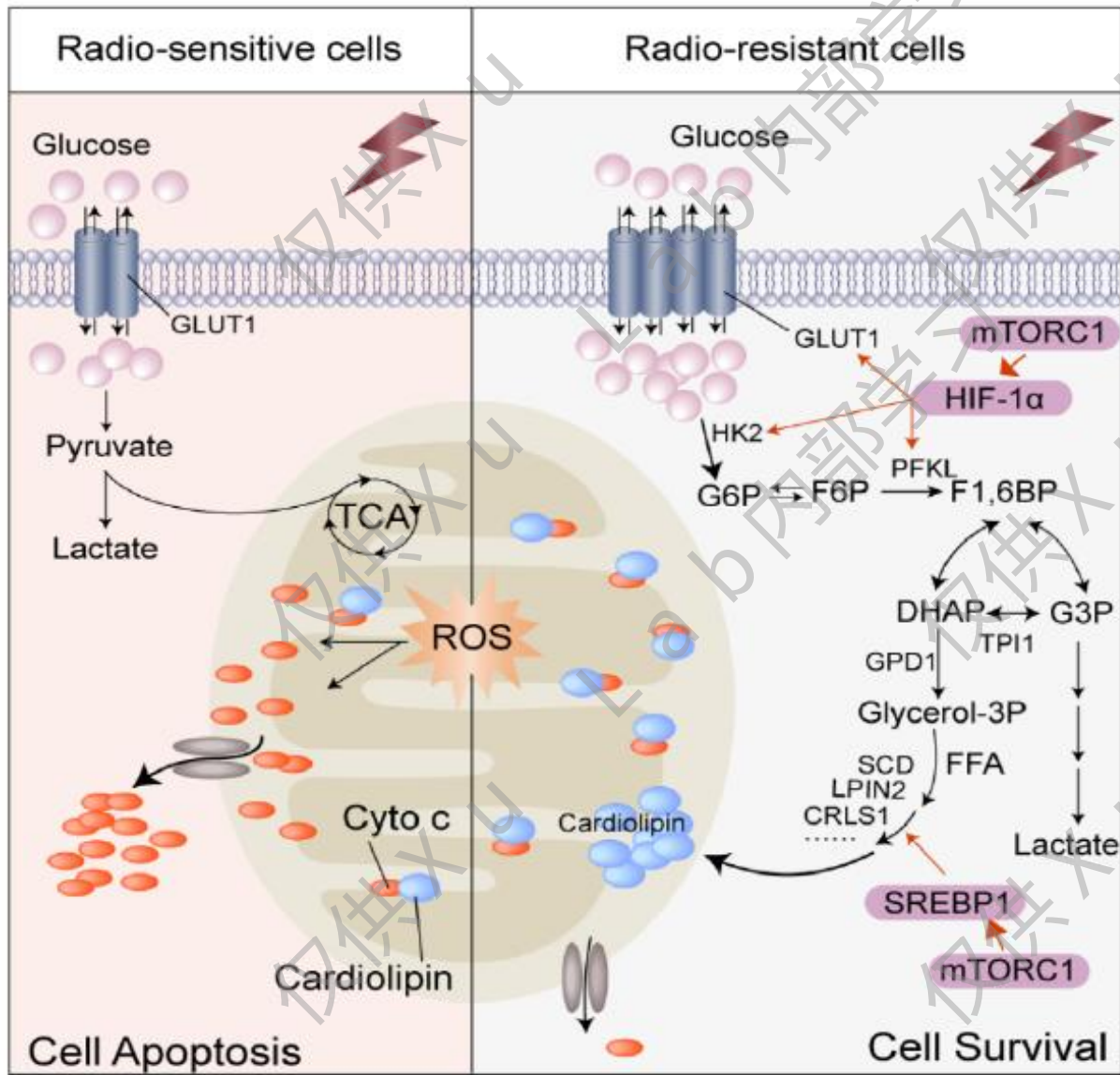
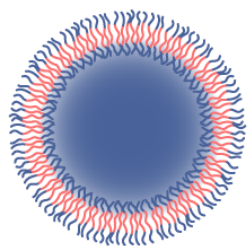


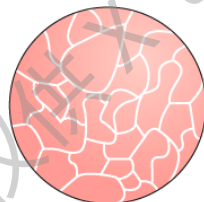




Polymeric



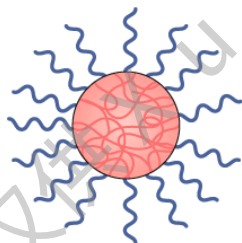
Polymersome



Dendrimer



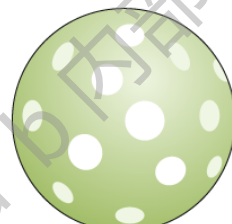
Polymer micelle



Nanosphere

- Precise control of particle characteristics
- Payload flexibility for hydrophilic and hydrophobic cargo
- Easy surface modification
- Possibility for aggregation and toxicity

Inorganic



Silica NP



Quantum dot



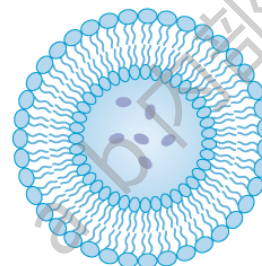
Iron oxide NP



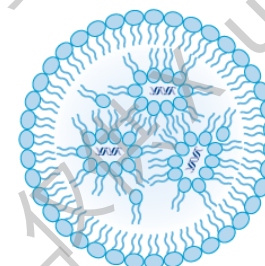
Gold NP

- Unique electrical, magnetic and optical properties
- Variability in size, structure and geometry
- Well suited for theranostic applications
- Toxicity and solubility limitations

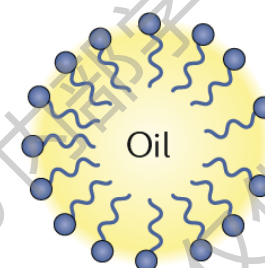
Lipid-based



Liposome



Lipid NP



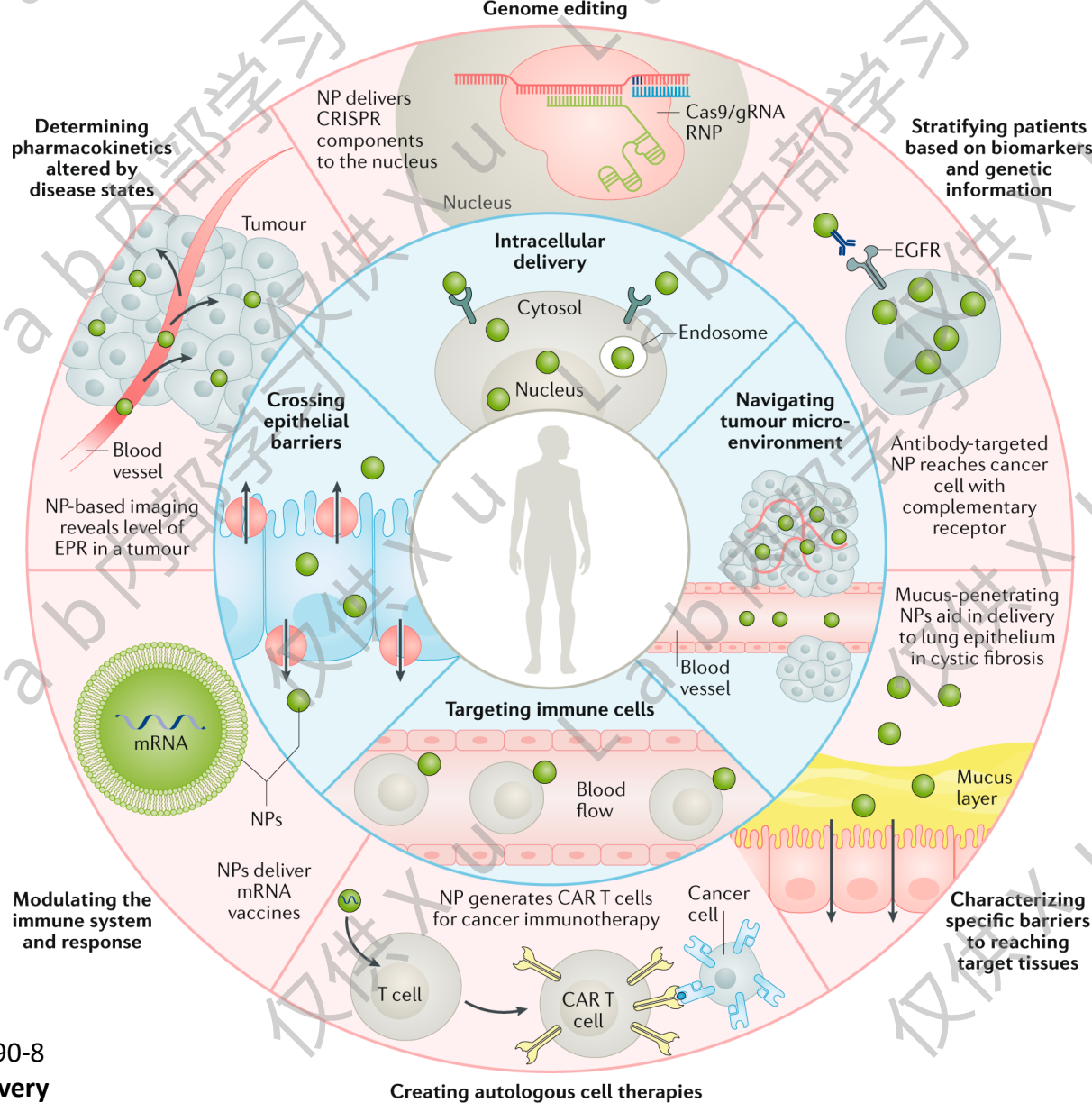
Emulsion

- Formulation simplicity with a range of physicochemical properties
- High bioavailability
- Payload flexibility
- Low encapsulation efficiency

<https://www.nature.com/articles/s41573-020-0090-8>

Engineering precision nanoparticles for drug delivery

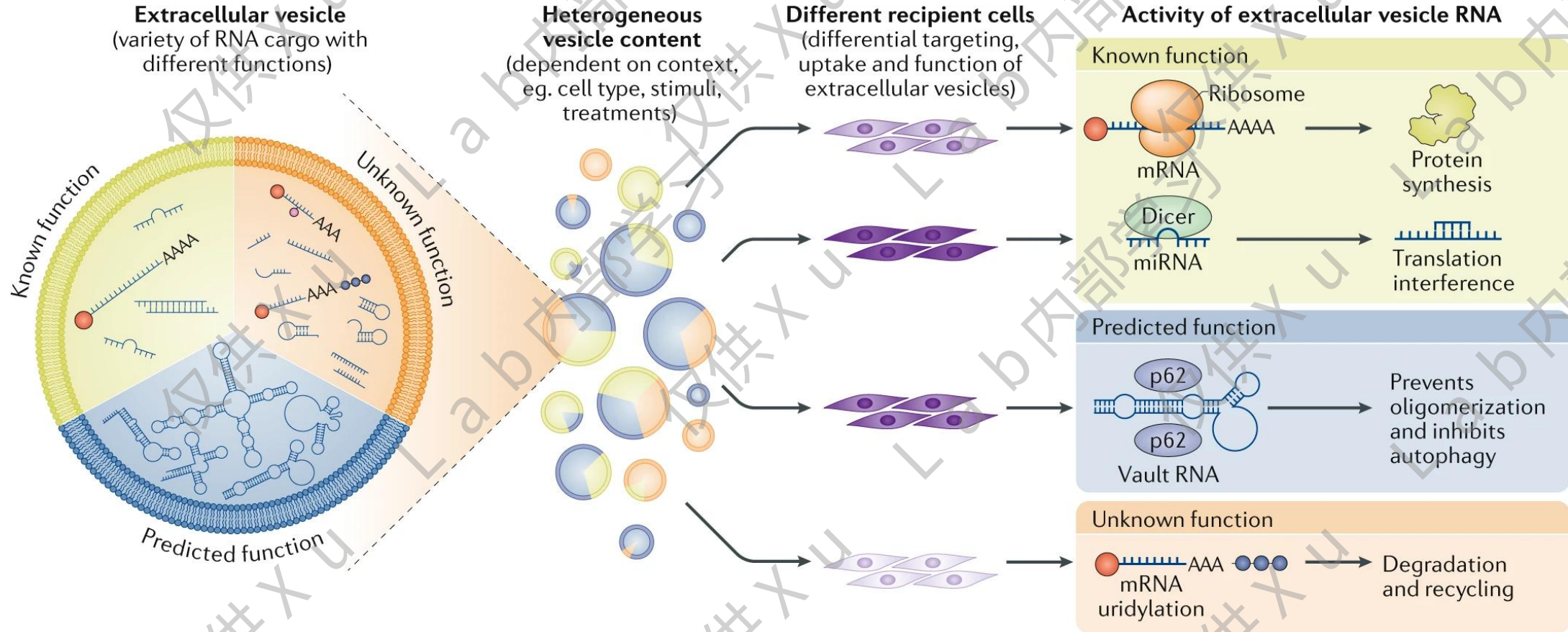
Nature Reviews Drug Discovery **volume 20**, pages101–124 (2021)



<https://www.nature.com/articles/s41573-020-0090-8>

Engineering precision nanoparticles for drug delivery

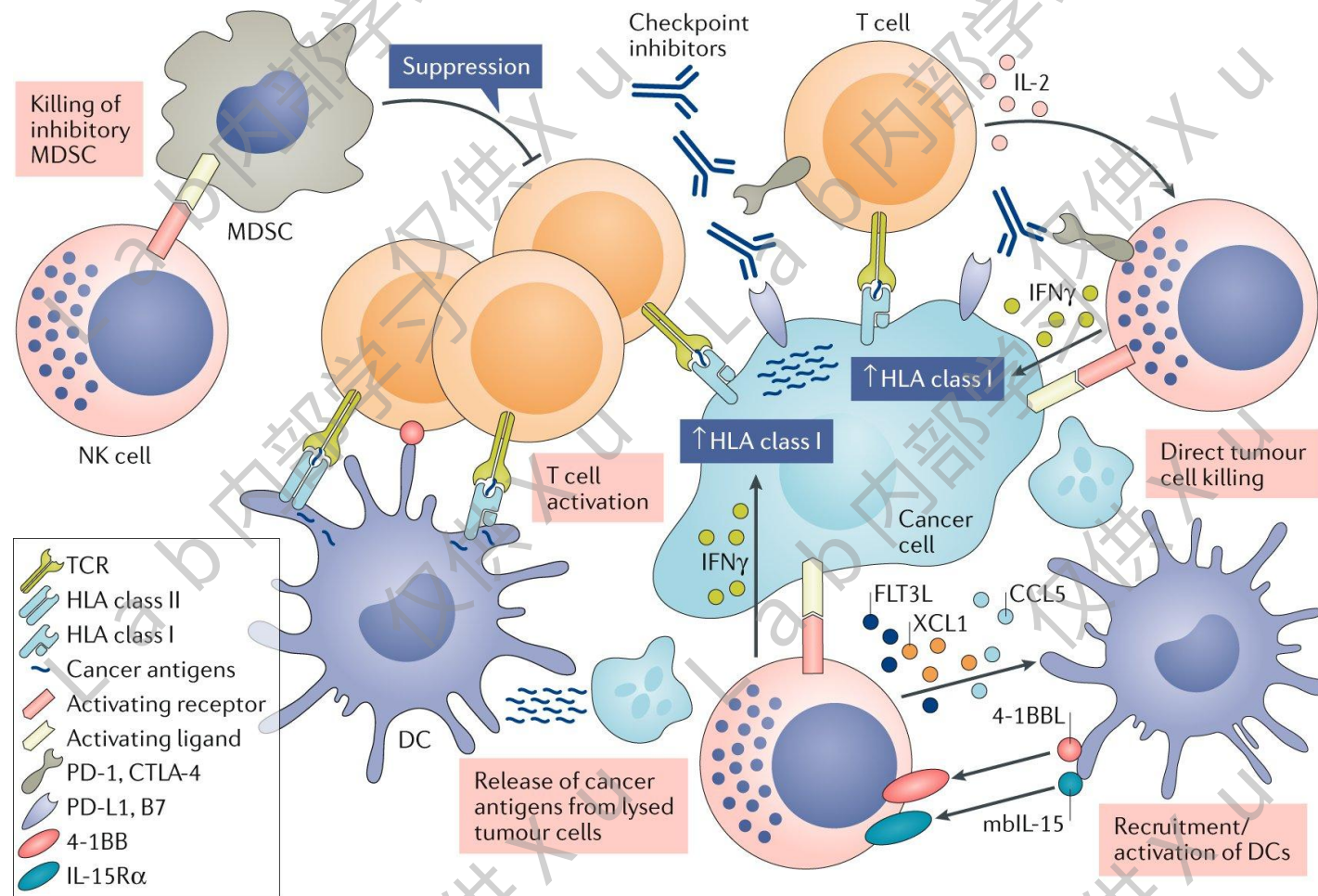
Nature Reviews Drug Discovery **volume 20**, pages101–124 (2021)

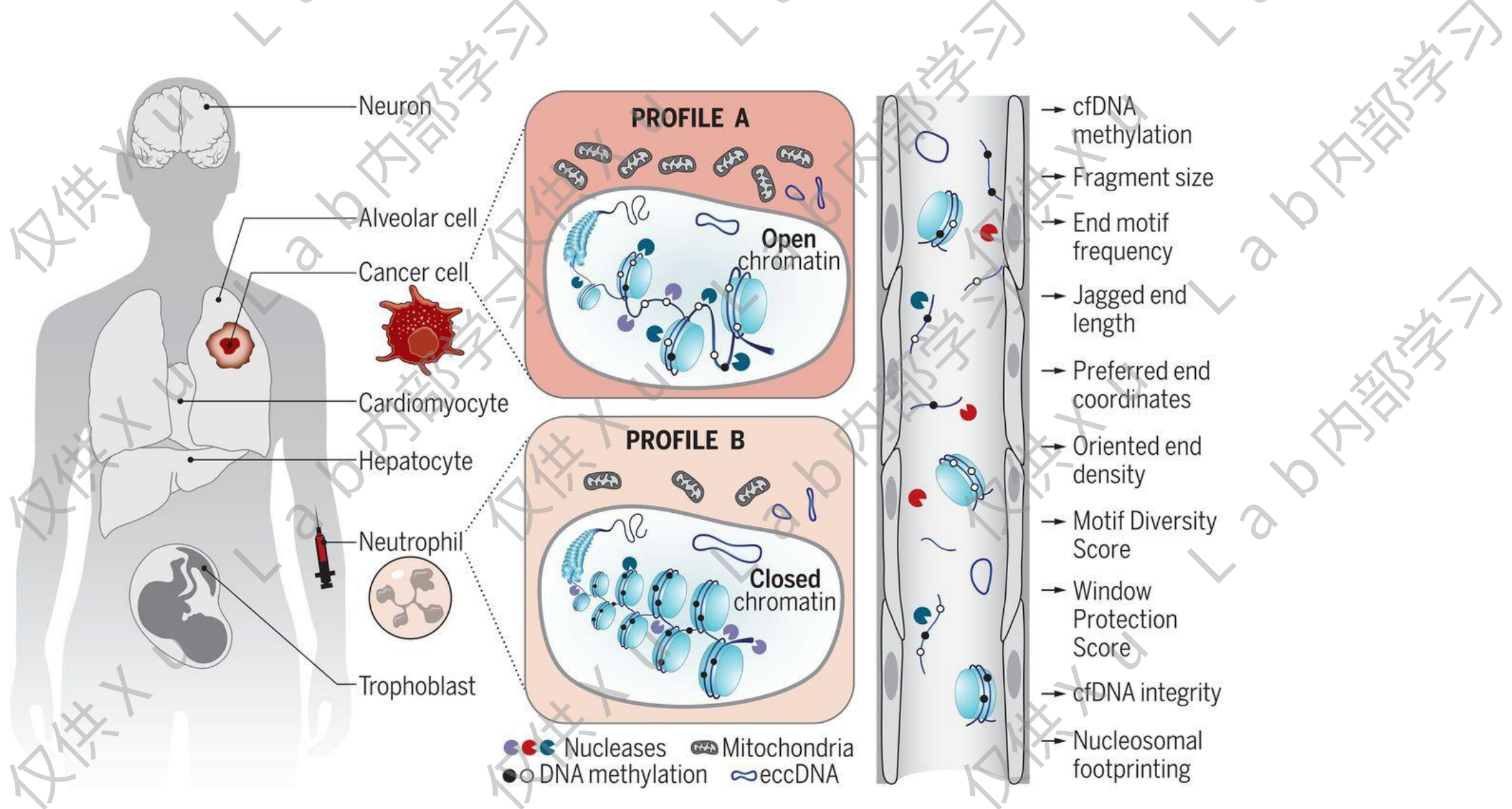


RNA delivery by extracellular vesicles in mammalian cells and its applications

Nature Reviews Molecular Cell Biology **volume 21**, pages585–606 (2020)

<https://www-nature-com.libproxy1.nus.edu.sg/articles/s41580-020-0251-y>

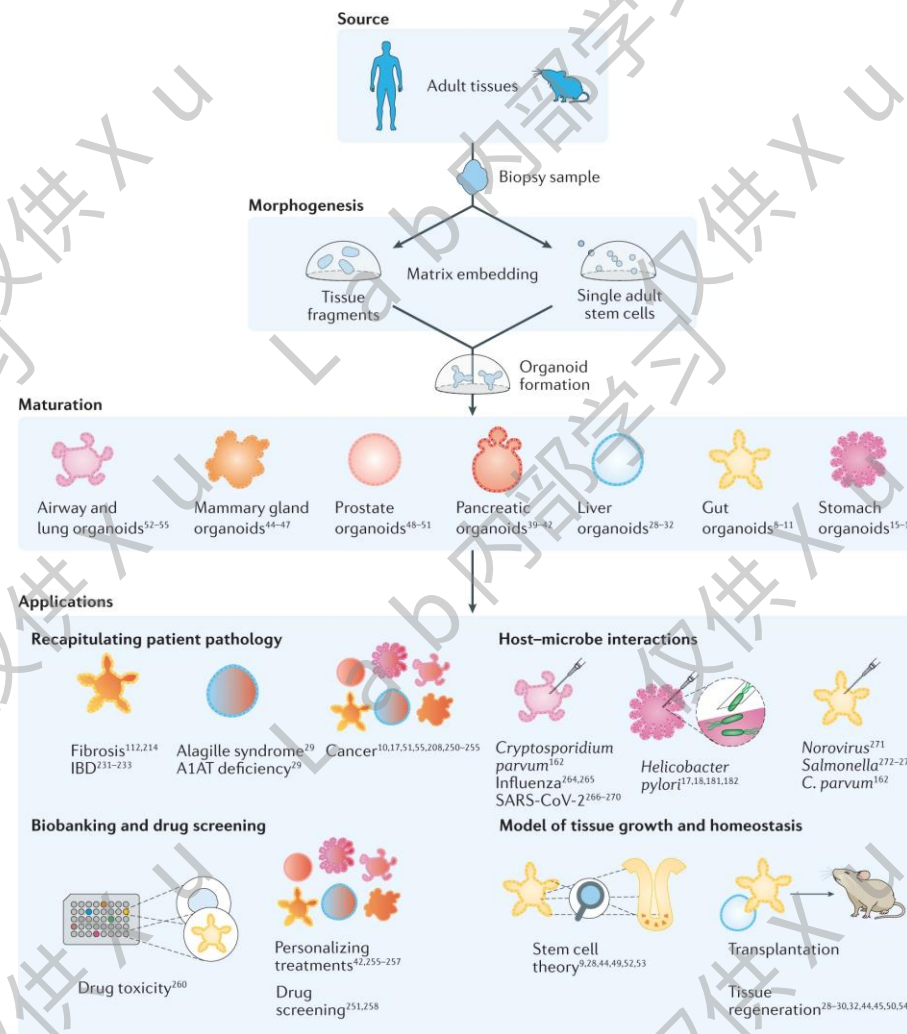
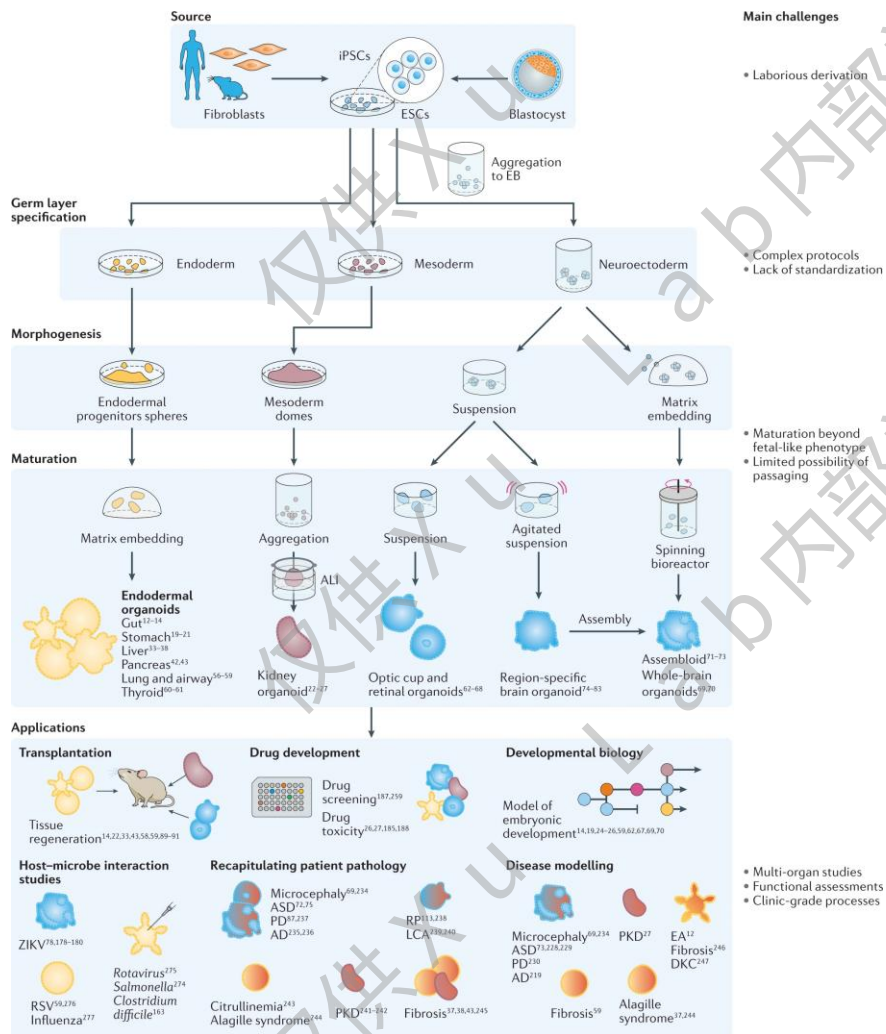




Science Magazine

<https://fclid.ly/3v3uhci>

Epigenetics, fragmentomics, and topology of cell-free DNA in liquid biopsies

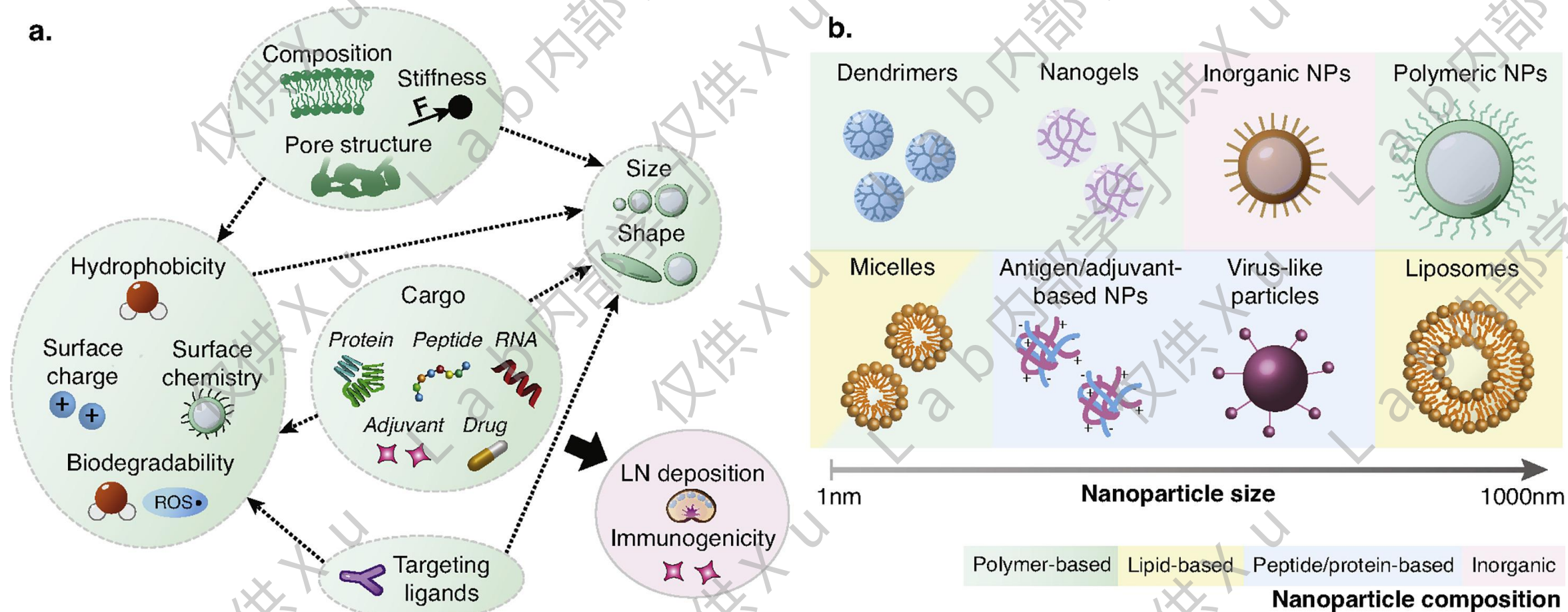


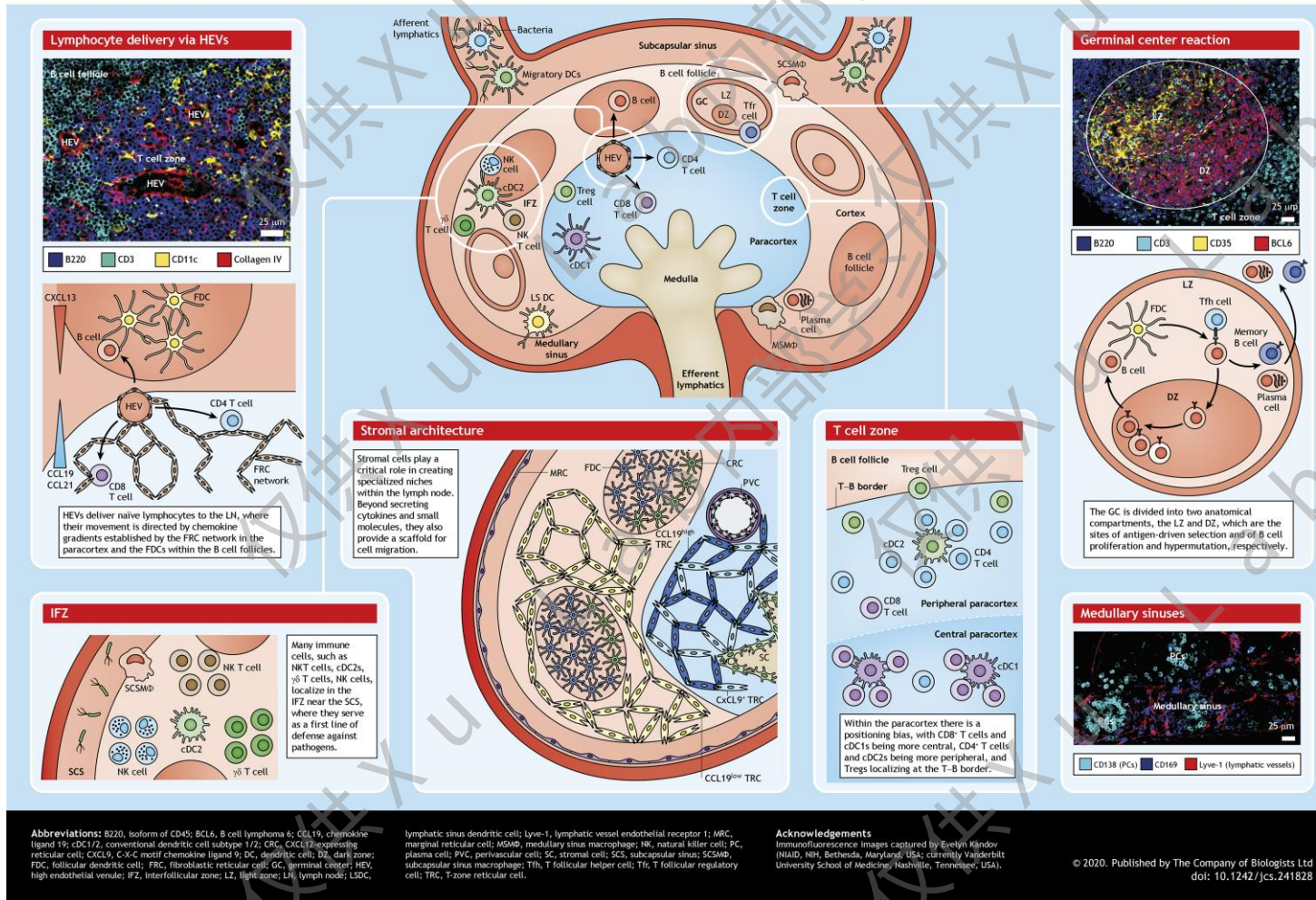
Nature Review Material

<https://www.nature.com/articles/s41578-021-00279-y#Fig1>

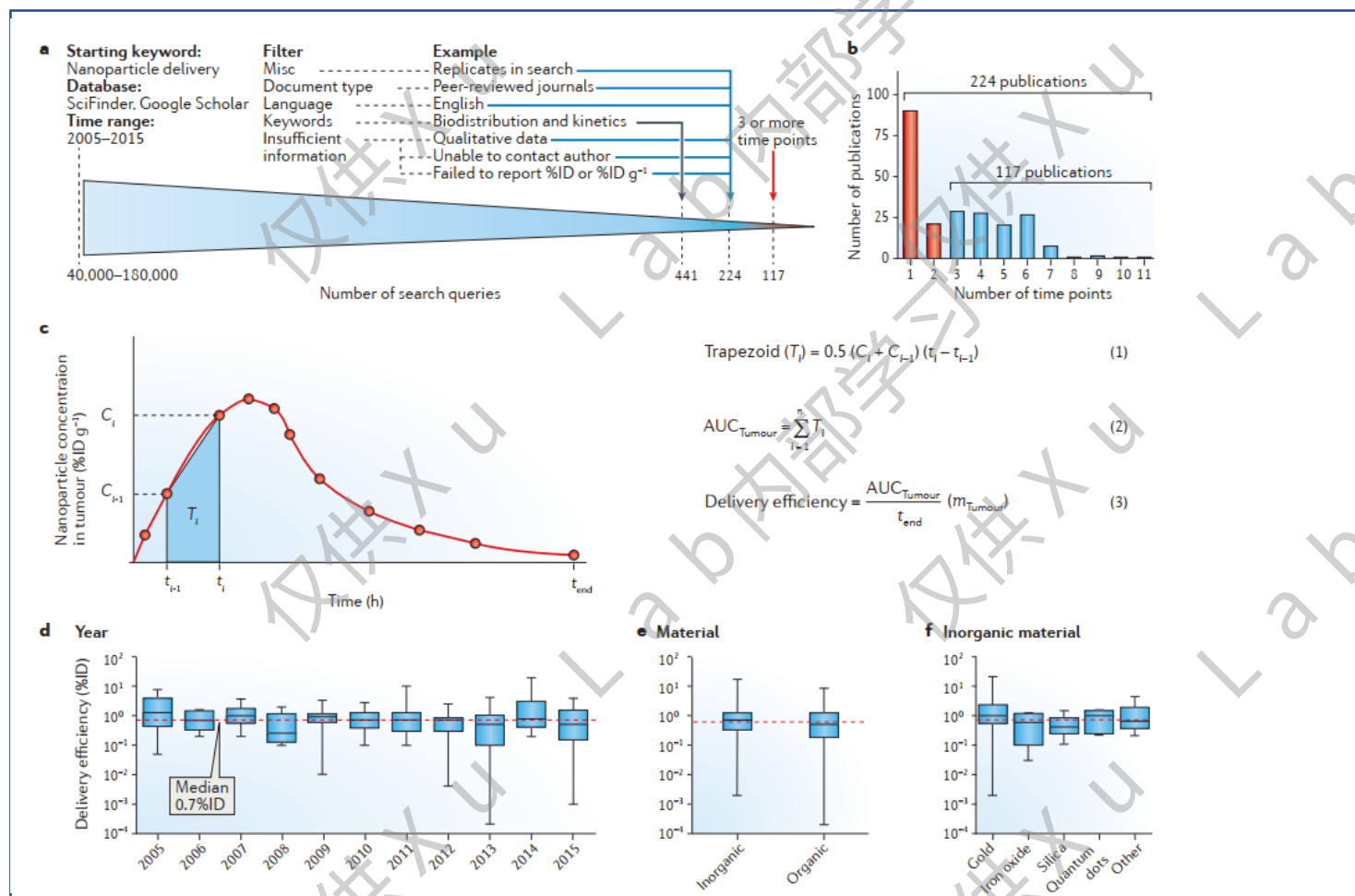
Engineering organoids

Different Stages for one technique





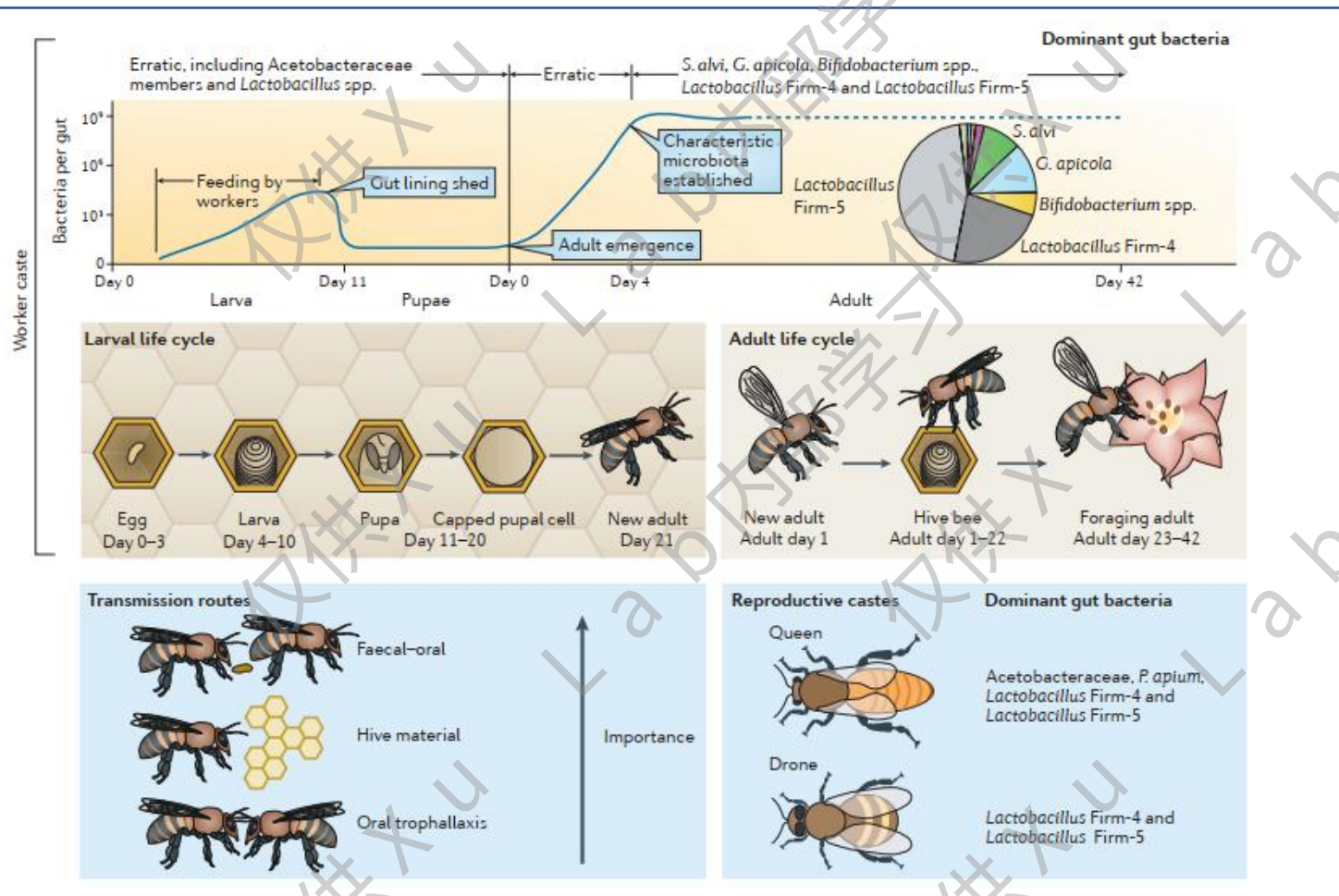
Detailed Analogy and structure of lymph node



Any comments if necessary

<https://www.nature.com/articles/natrevmats201614>

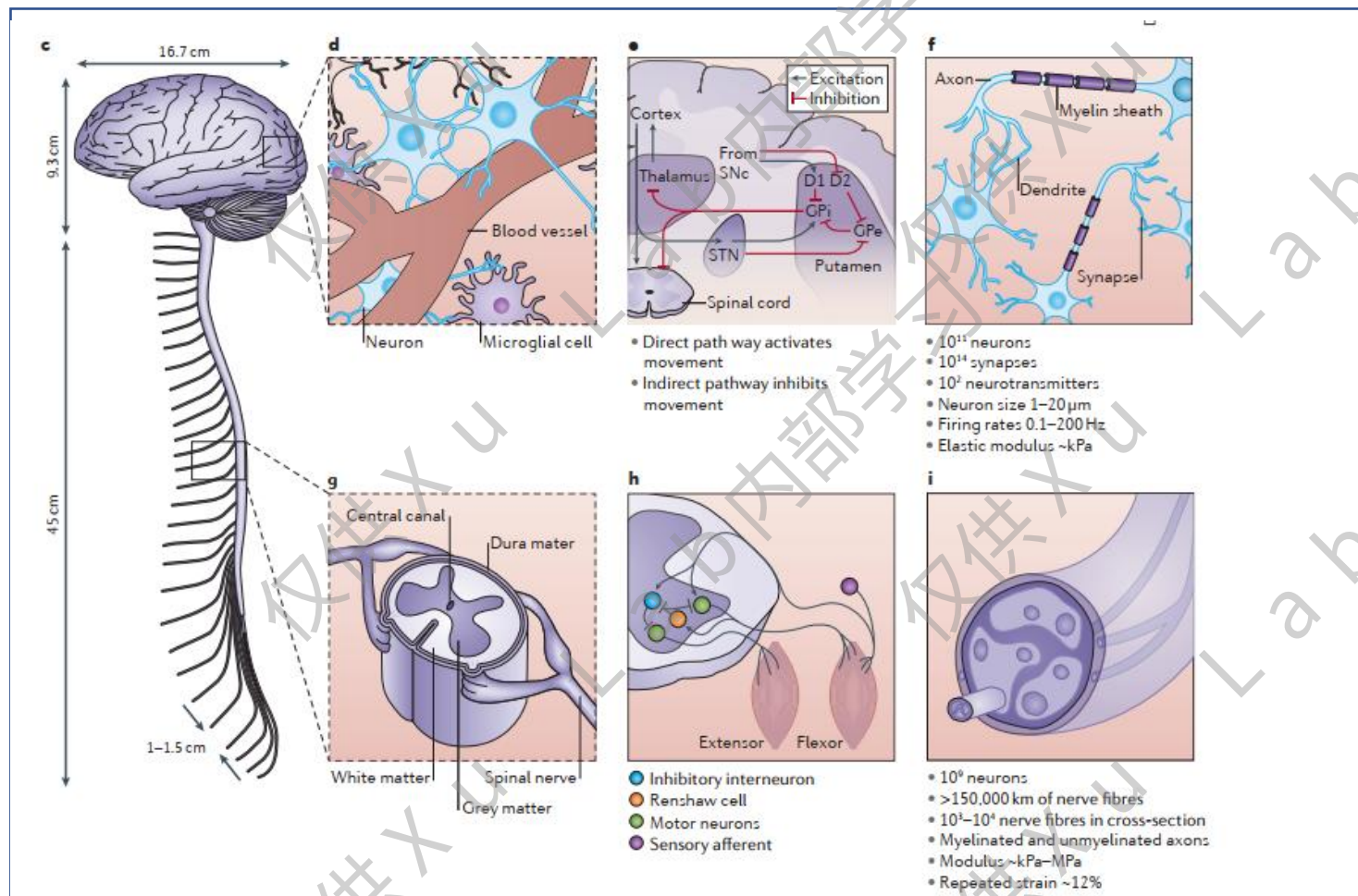
15th Aug 2021



Any comments if necessary

<https://www.nature.com/articles/nrmicro.2016.43>

15th Aug 2021

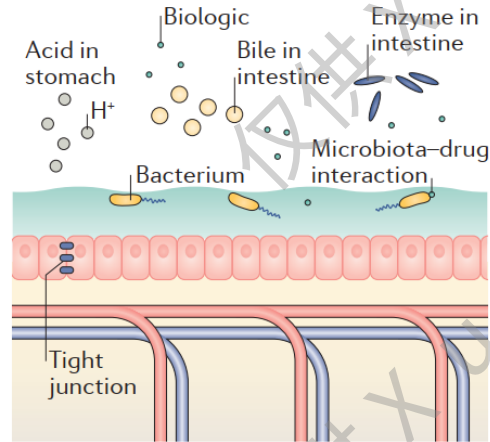


Any comments if necessary

<https://www.nature.com/articles/natrevmats201693>

15th Aug 2021

a Delivery challenges



b New approaches

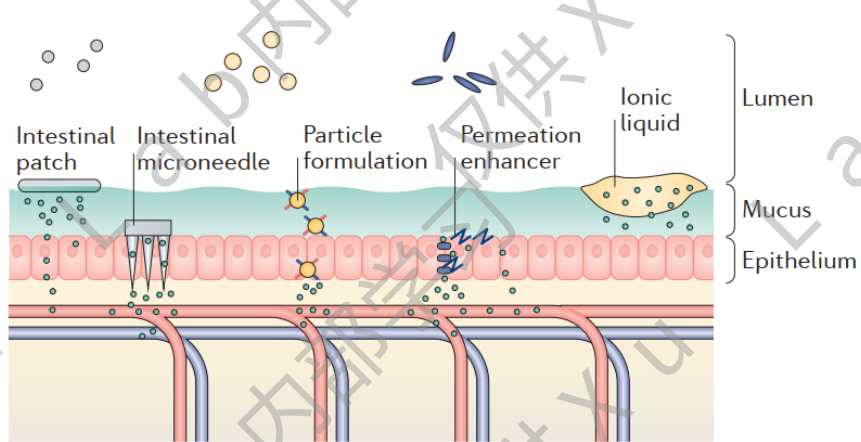
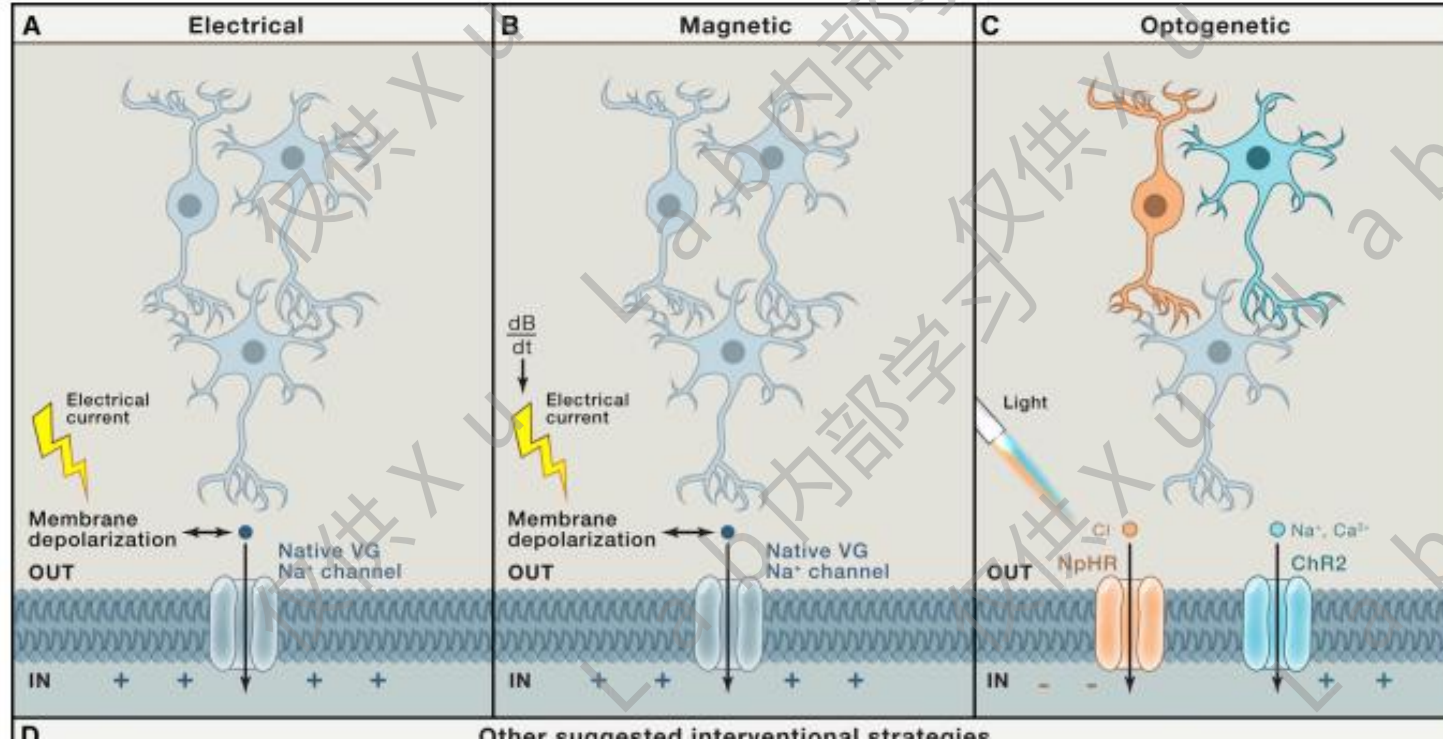


Figure 3 | **Oral barriers, products and new approaches.** **a** | Delivery barriers and microenvironmental challenges that limit biologic absorption via the oral route. **b** | Approaches to overcome these barriers and microenvironmental challenges. Intestinal patches improve absorption by enhancing mucoadhesion and subsequently diffusion. Intestinal microneedles can be used to facilitate absorption by physically breaching the epithelium. Particle formulations can improve transport by including intestine-targeting ligands, mucoadhesive coatings or mucopenetrative coatings.

Any comments if necessary

<https://www.nature.com/articles/nrd.2018.183>

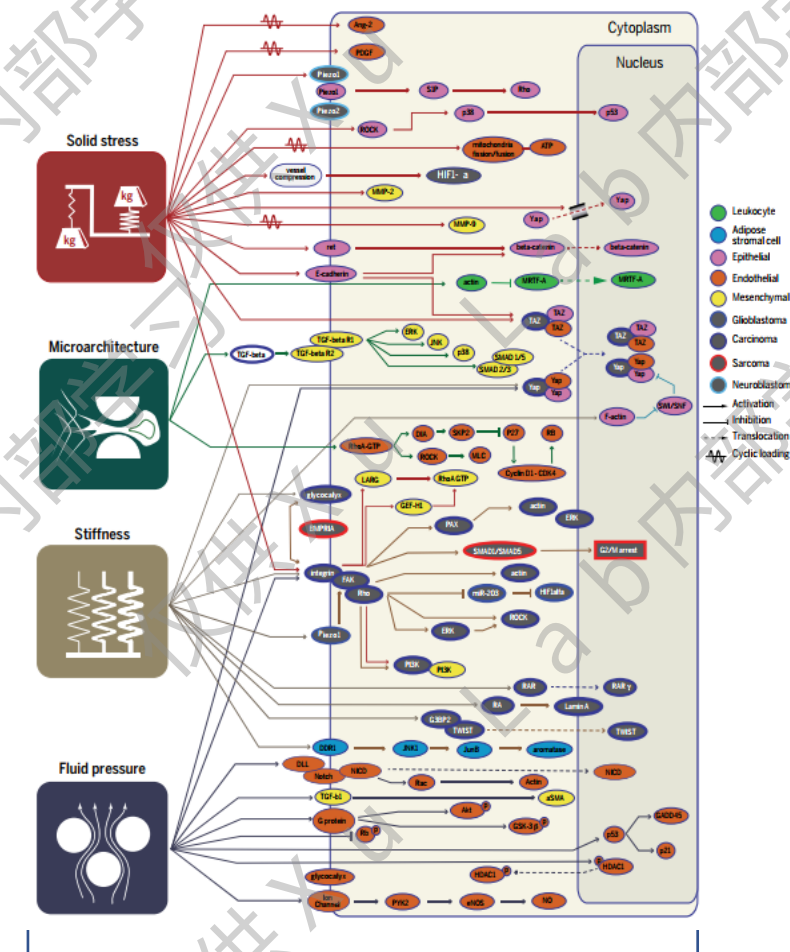
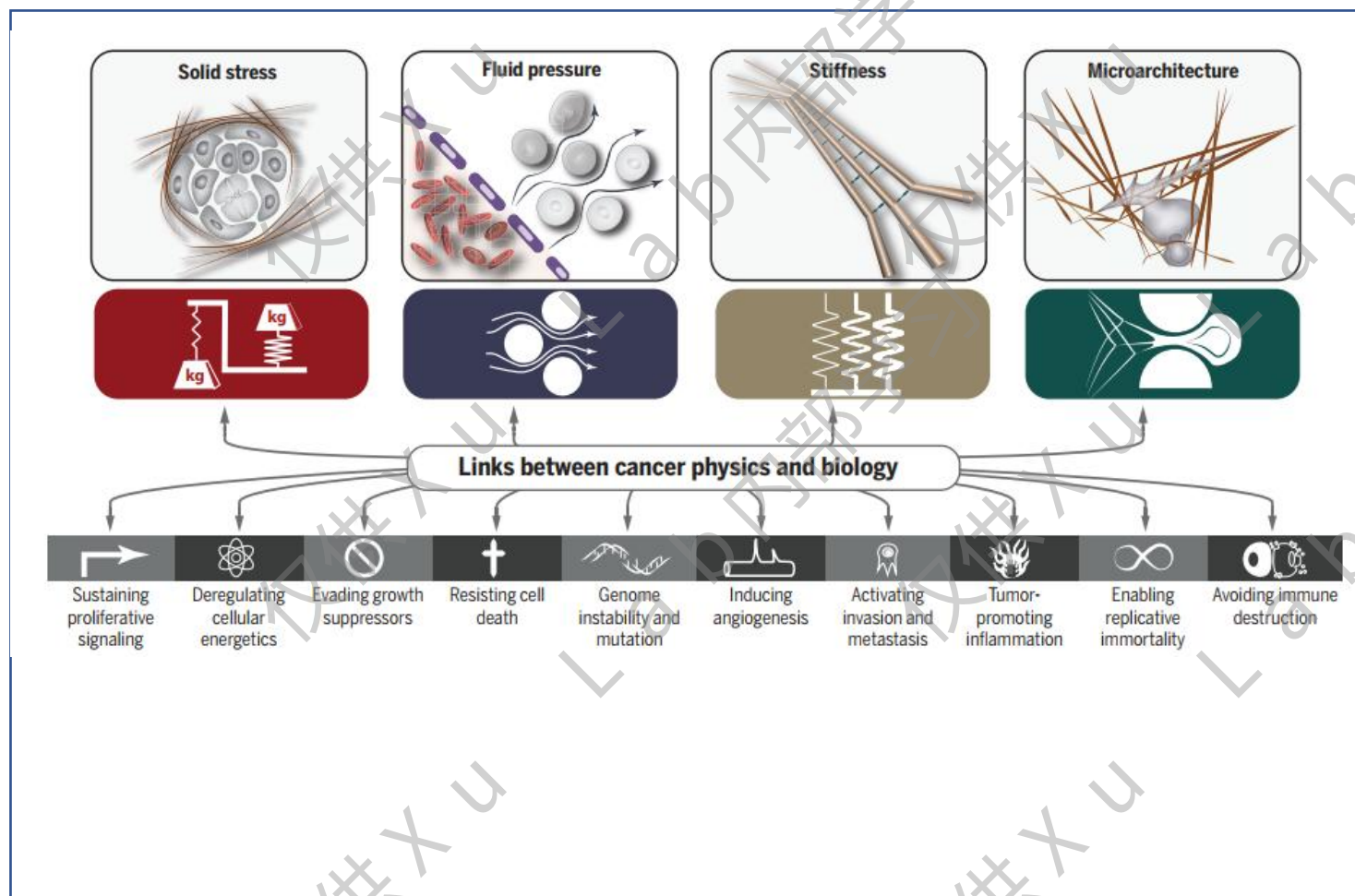
15th Aug 2021



Any comments if necessary

<https://pubmed.ncbi.nlm.nih.gov/27104976/>

15th Aug 2021



<https://science.sciencemag.org/content/370/6516/eaaz0868>

21st Aug 2021

CELL FORCE IS A HUGE PROBLEM

FIBROSIS

IDIOPATHIC PULMONARY FIBROSIS
LIVER AND CARDIAC FIBROSIS

Role of force: Too high—Increased myofibroblast force promotes fibrotic remodeling

Therapeutic strategy: Inhibit myofibroblast contractility and transdifferentiation into contractile cells

US market: ~1–2% prevalence | \$2.5B market

CARDIOVASCULAR

HYPERTENSION

Role of force: Too high—Constricted vascular cells narrow blood vessels and lead to greater resistance to bloodflow

Therapeutic strategy: Relax airway SMCs to increase lumen diameter

US market: 76mm patients | \$46B annual revenue

REPRODUCTIVE

PRE-TERM LABOR (FEMALE)

Role of force: Uncontrolled—Sudden, uncontrolled uterine contractions lead to early delivery

Therapeutic strategy: Arrest contractions to delay birth

US market: 1 in 5 births | \$50k per preterm infant

IMMUNE SYSTEM

IMPAIRED PHAGOCYTOSIS IN
AUTOIMMUNITY/ INFLAMMATION

Role of force: Too low—Phagocytes unable to clear apoptotic debris or foreign pathogens

Therapeutic strategy: Augment and restore effective levels of force

US market: 1–2% prevalence | \$100B annual revenue

MIGRAINE

Role of force: Too low—Increased bloodflow due to sudden cranial vasodilation leads to pain response

Therapeutic strategy: Protect cranial arteries from CGRP-induced relaxation

US market: 15% prevalence | \$17B annual revenue

NEUROVASCULAR

GLAUCOMA

OCULAR

Role of force: Too high—The highly contractile trabecular meshwork creates high resistance to aqueous humor outflow leading to high intraocular pressure and nerve damage

Therapeutic strategy: Relax the trabecular meshwork cells to increase pore size and decrease outflow resistance leading to greater drainage from the eye

US market: ~3.5% of adults over age 40

ASTHMA

RESPIRATORY

Role of force: Too high—Obstruction of airflow due from increased airway SMC contractility

Therapeutic strategy: Relax airway SMCs to increase lumen diameter

US market: 26mm patients | \$56B annual revenue

SPASTIC BLADDER/ INCONTINENCE

UROLOGY

Role of force: Too low—Sudden, uncontrolled bladder spasms lead to severe discomfort and loss of voluntary bladder control

Therapeutic strategy: Relax bladder SMCs to alleviate condition

US market: 8% prevalence | \$3B annual revenue

METASTASIS AND TUMOR TENSION

ONCOLOGY

Role of force: Too low—Enables invasion and metastasis and prevents therapeutics from penetrating tumors

Therapeutic strategy: Inhibit force production in cancer cells to prevent invasion and enhance drug delivery efficacy

US market: 1.7M new patients, annually | \$147B market

ments if necessary

21st Aug 2021

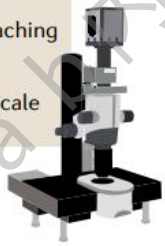
a Epifluorescence and stereomicroscopy

- Staining with fluorophores, dyes or HRP-DAB immunohistochemistry
- Instant imaging
- Low resolution
- Poor optical penetration



b Light-sheet fluorescence microscopy

- Illumination in a single plane
- Reduced photobleaching
- Rapid imaging
- Whole organs and entire centimetre-scale organisms



c Confocal microscopy

- Whole-sample illumination and pinhole exclusion
- High optical resolution and magnification
- Low working distance
- Slow imaging

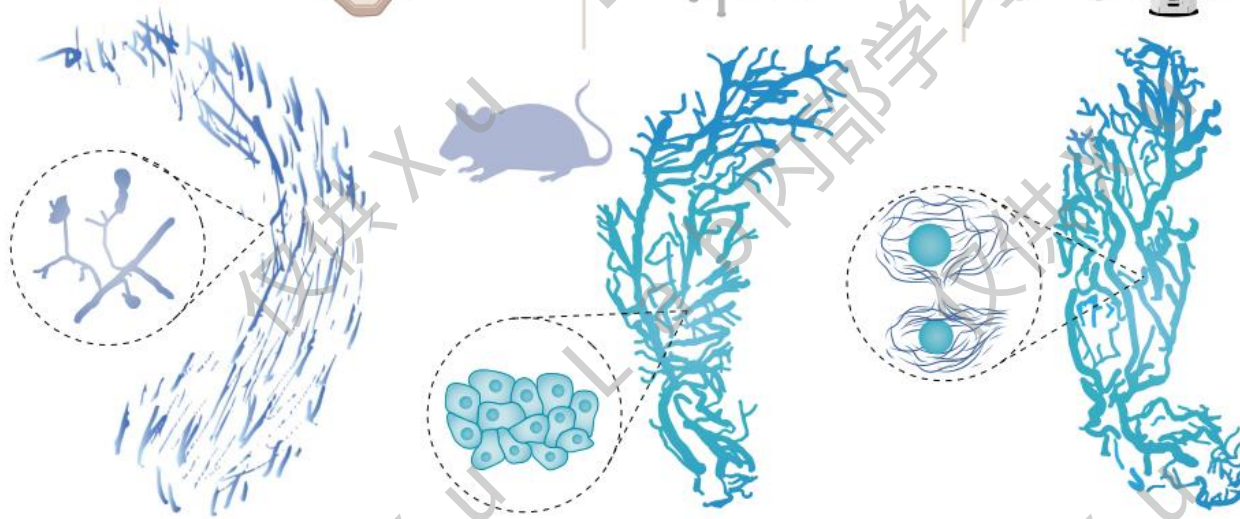
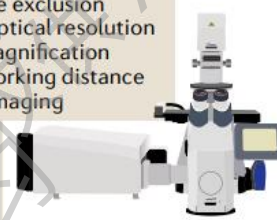
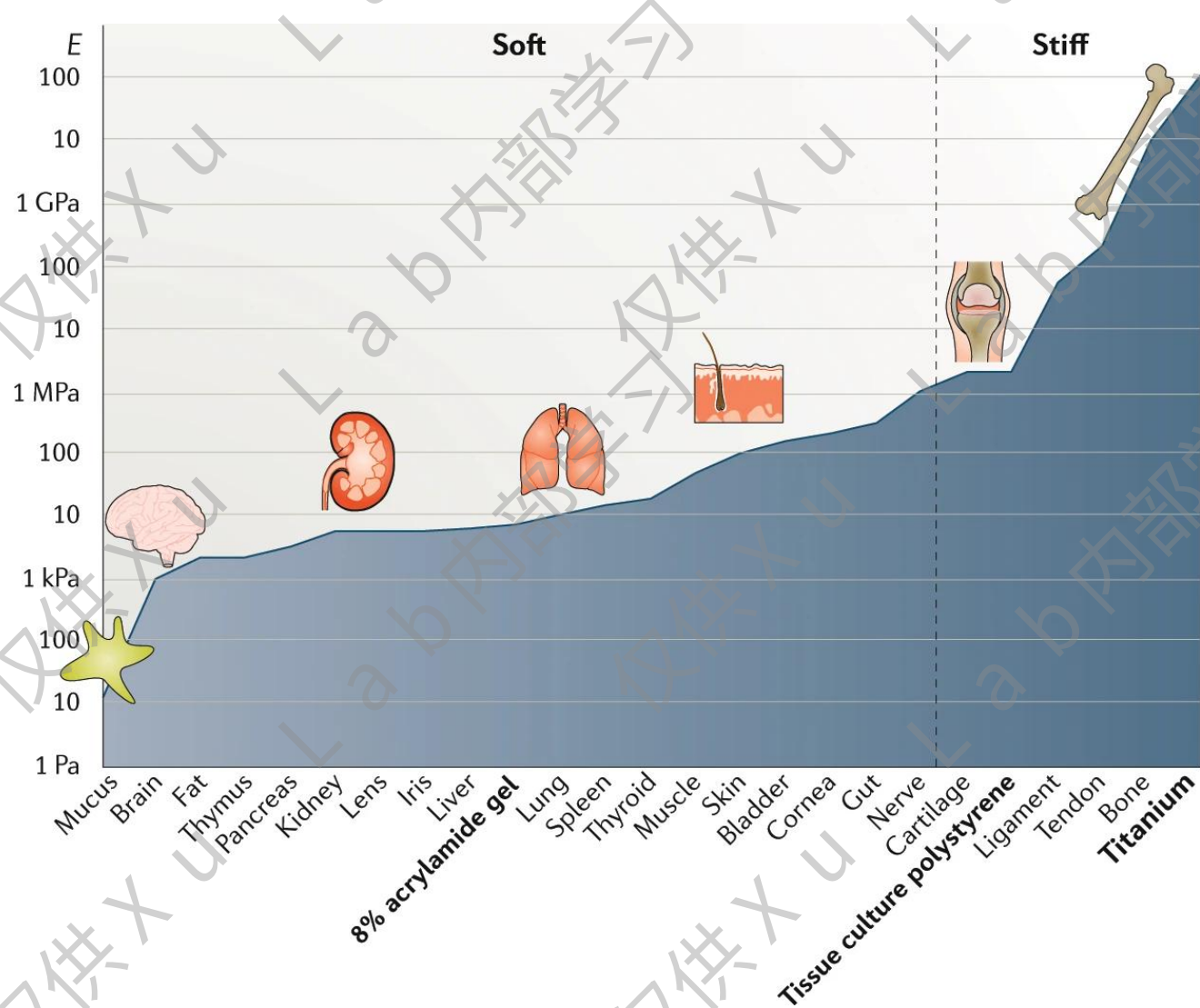


Fig. 2 | **Imaging approaches for transparent tissues.** a | Properties of epifluorescence and stereomicroscopy for cleared tissue (top); schematic of a whole cleared murine mammary gland and details of its ducts under the epifluorescence

Any comments if necessary

<https://www.nature.com/articles/s41568-021-00382-w>

21st Aug 2021

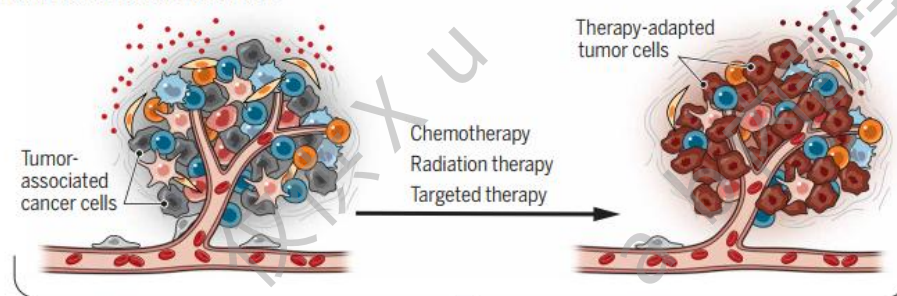


<https://www.nature.com/articles/s41578-019-0169-1?proof=t>

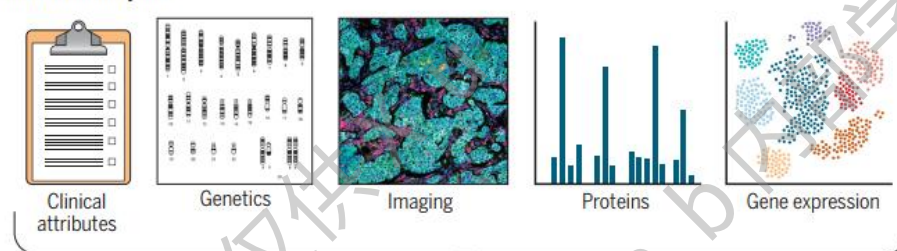
The stiffness of living tissues and its implications for tissue engineering

6 Oct 2021

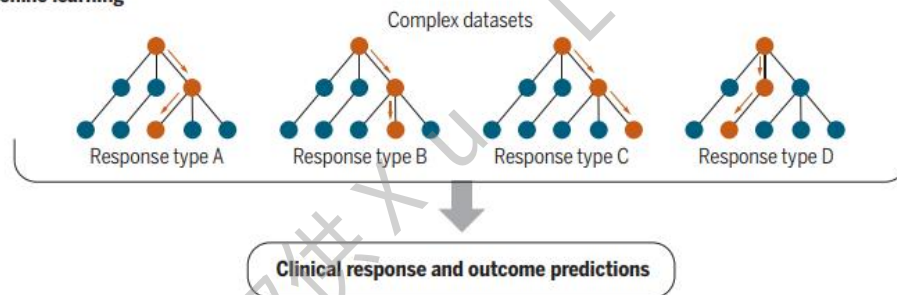
Dynamic tumor microenvironment



Multimodal techniques



Machine learning

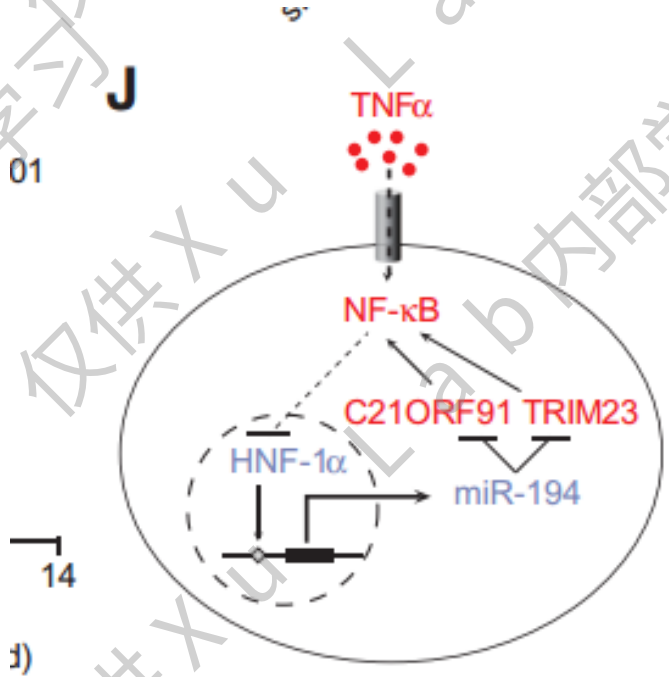
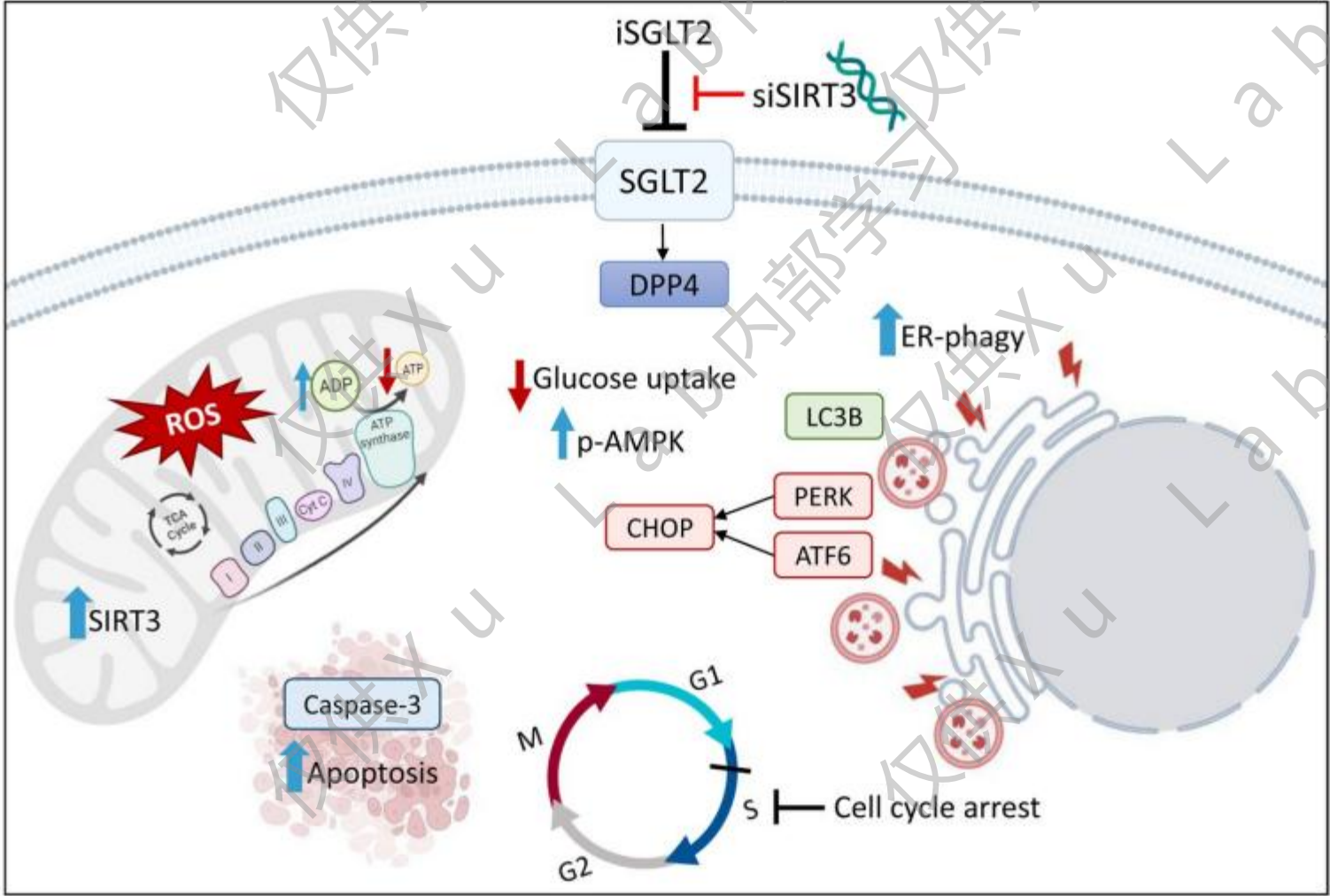


<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC8177696/>

Decoding the dynamic tumor microenvironment

6 Oct 2021

Graphical Abstract



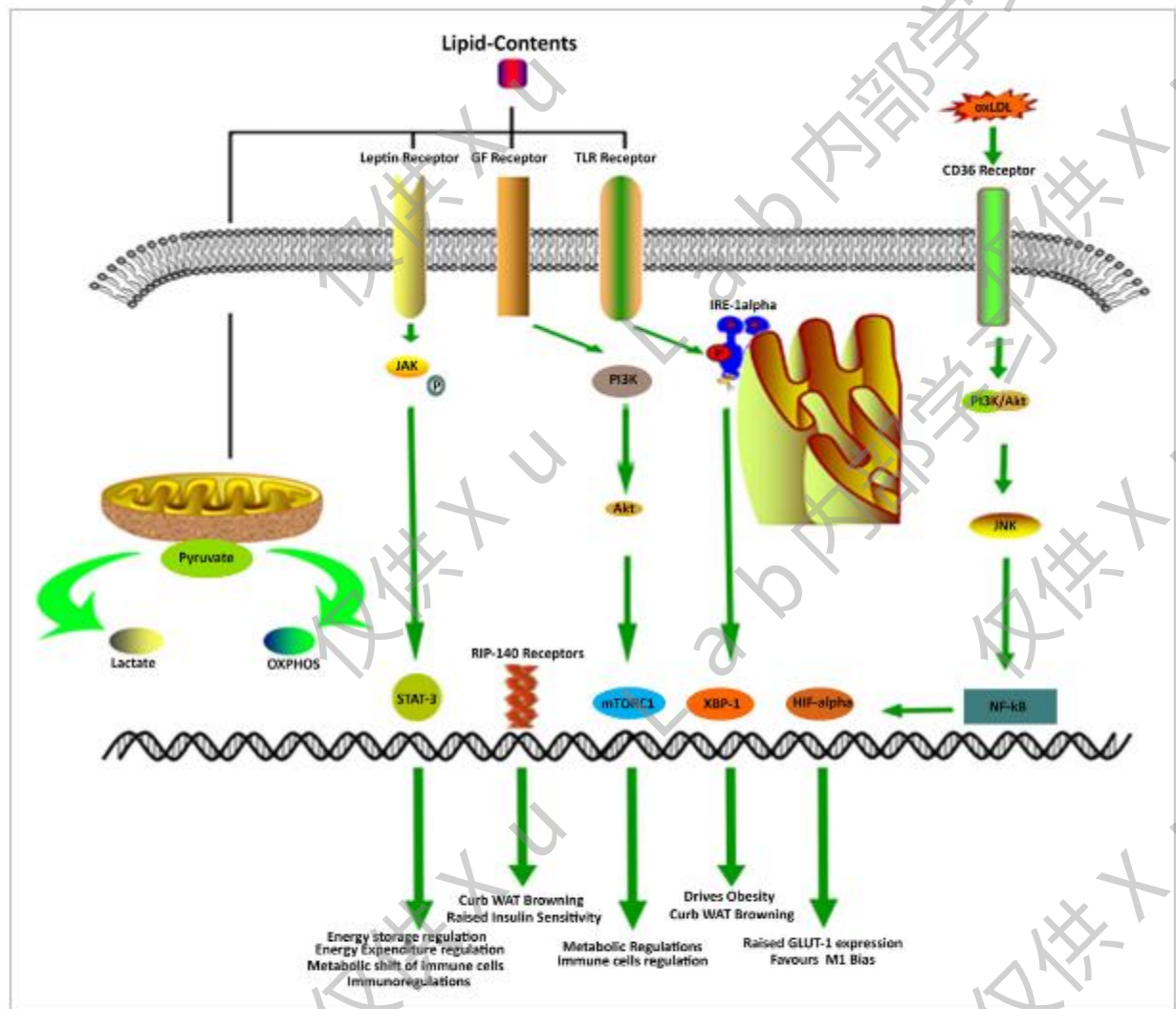
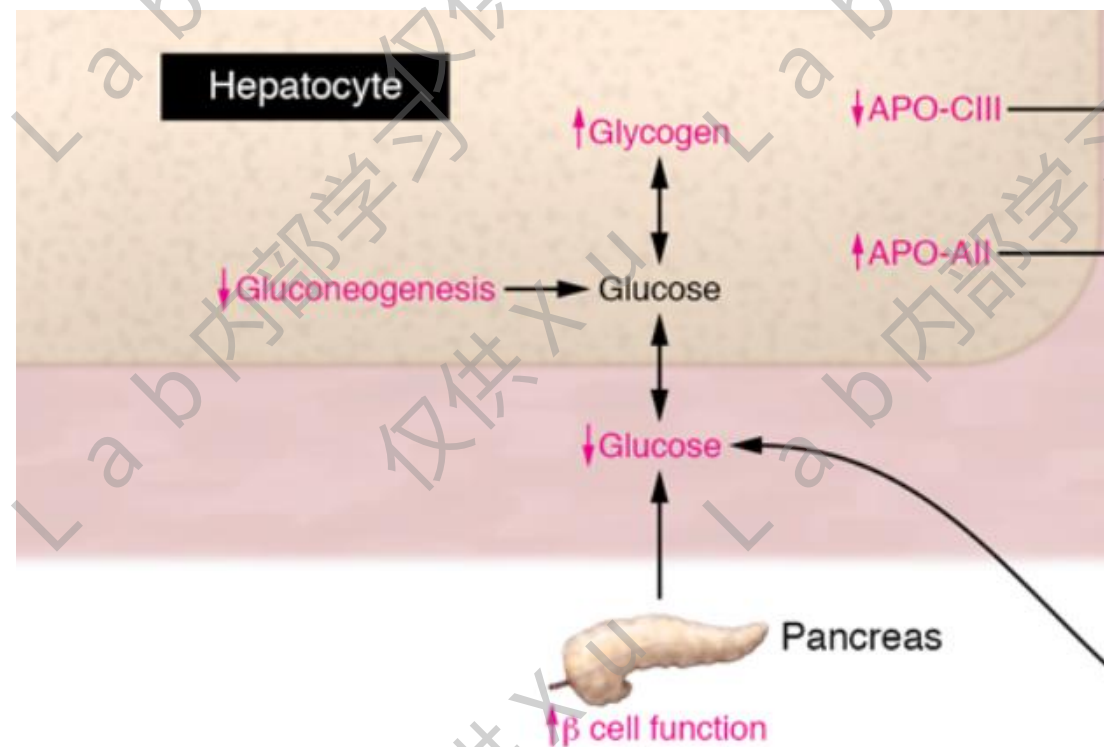


Figure 1

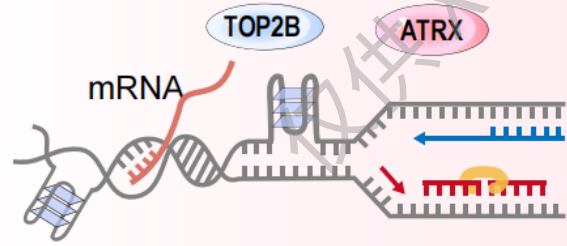
[Open in figure viewer](#) | [PowerPoint](#)

Molecular interactions in Immunometabolism governing lipid metabolic disorders. Induction of Leptin receptor, GF receptors, TLR receptors and CD36 receptors lead to the activation of STAT3, mTORC1, XBP-1 (through IRE-1α) and HIF-1α that steer to interference in lipid

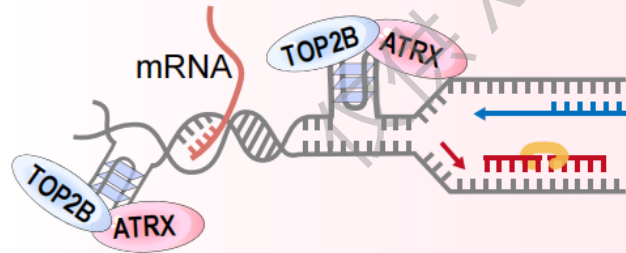


Graphical abstract

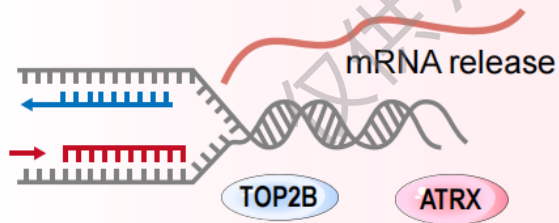
Normal glioma cell



G4s obstruct replication and transcription

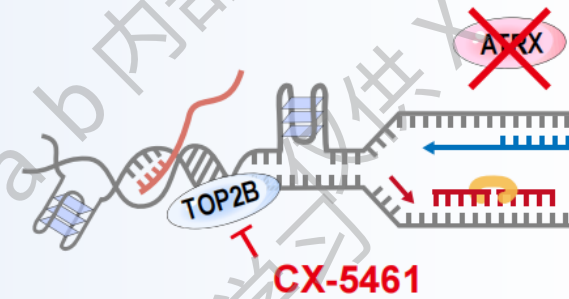


TOP2B-ATRAX complex resolves G4

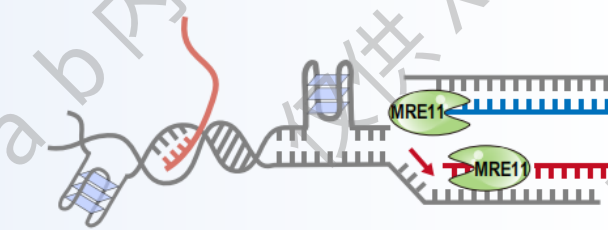


Replication and transcription proceed

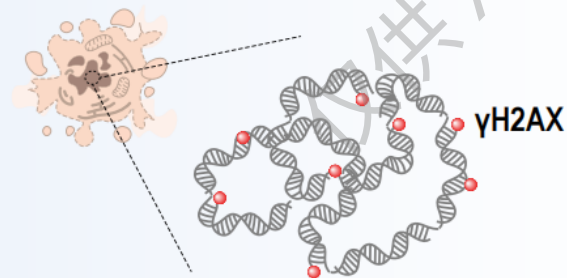
ATRX-deficient glioma cell



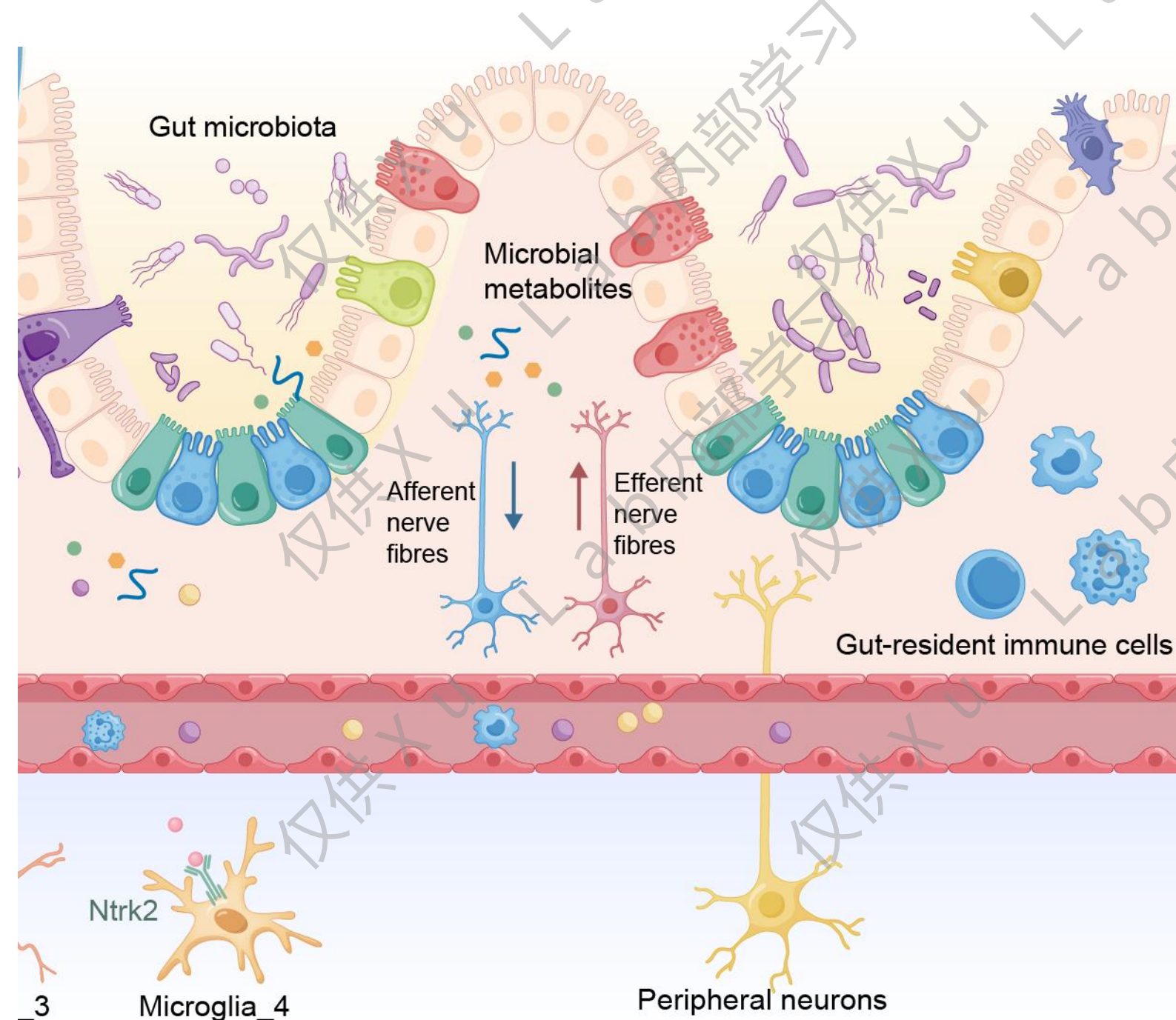
CX-5461 induces G4 accumulation



Transcription block, replication fork collapse



DSBs accumulation, cell death

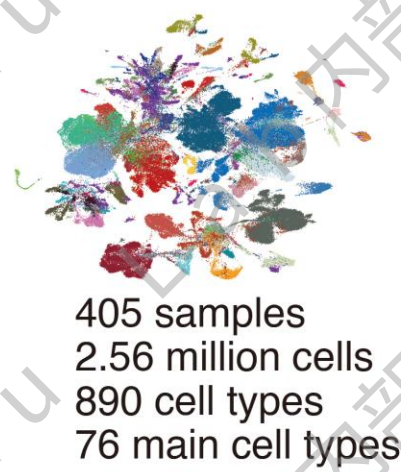
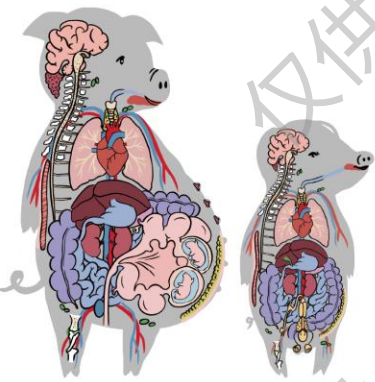


仅供X u L a b 内部学习

"All-from-one" cell atlas from a pig family

Mother
119 tissue

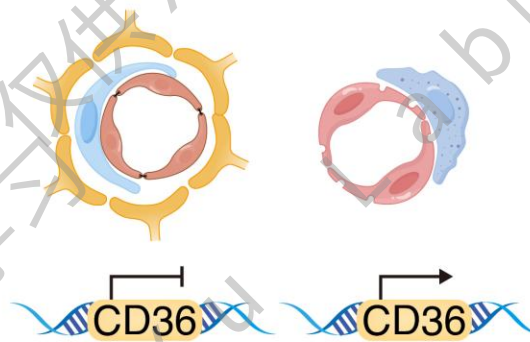
Son
115 tissue



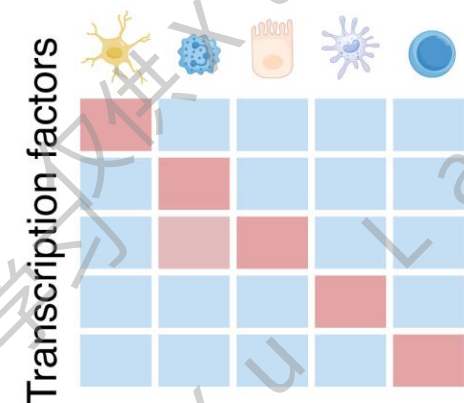
Endothelial cell diversity

Brain

Peripheral organs

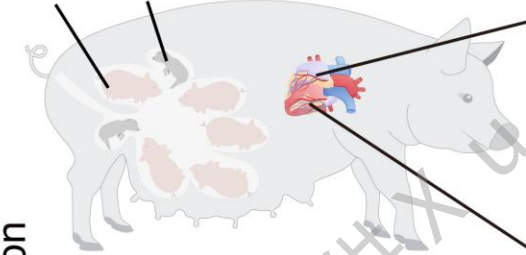


Cell type-specific TF



Cardiac cell remodeling in pregnancy

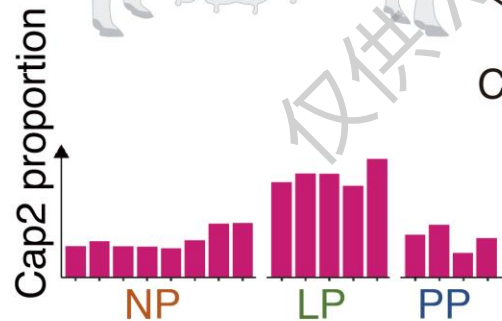
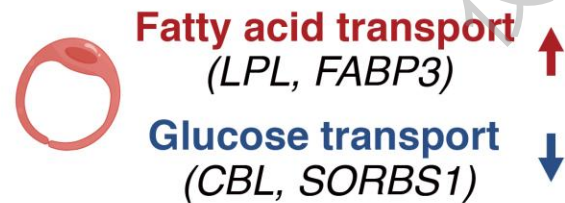
NW GR



Cardiomyocytes



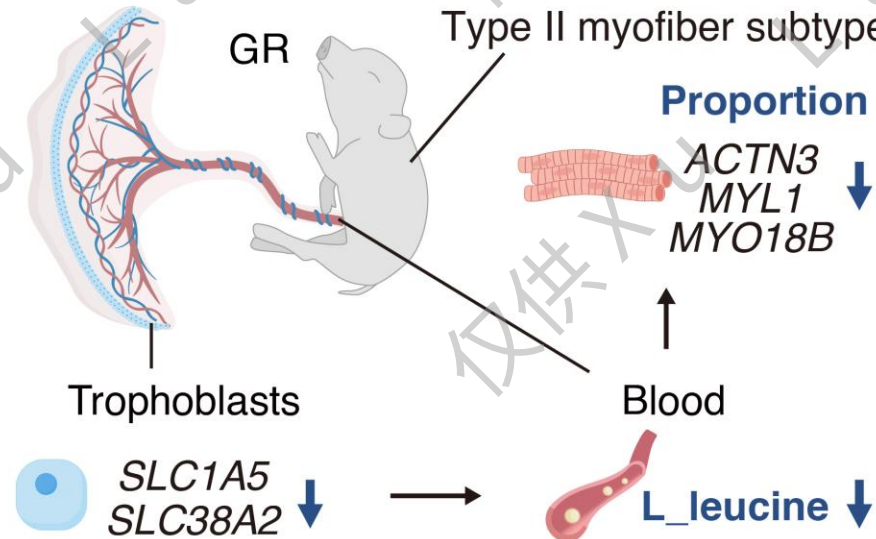
Cap2 Endothelial cell

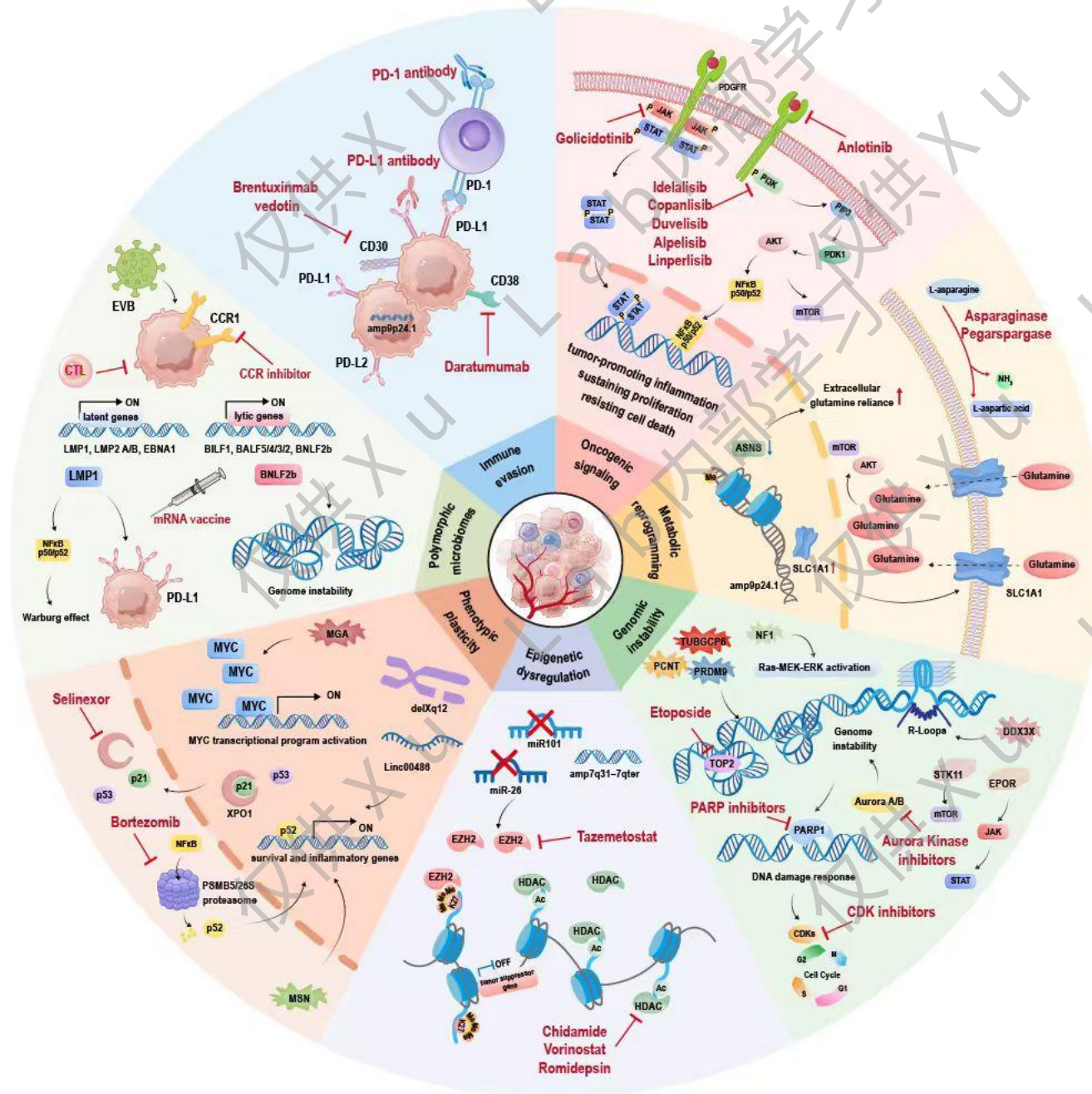


Cellular changes in fetal growth restriction

GR

Type II myofiber subtype





仅供内部学习